Proactive Heating Control Using Forecasted Data with Q-learning

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***Abstract*—The growing focus on energy efficiency and sustainability in building management has highlighted the need for advanced heating control systems. This report investigates the implementation of a proactive Q-learning algorithm for residential heating control, utilizing forecasted weather data to enhance both energy efficiency and occupant comfort. By comparing a traditional thermostat model with the Q-learning-based system, the study demonstrates significant improvements in maintaining stable indoor temperatures while reducing energy costs. The Q-learning model, characterized by its continuous learning process, outperformed the thermostat model by optimizing heating adjustments in real-time, resulting in a 9.5% reduction in heating costs. Additionally, the study underscores the potential for further refinement of the Q-learning algorithm to balance the frequency of adjustments with the longevity of heating system components.**

***Index Terms*—Proactive Heating Control, Retroactive Heating Control, Q-learning Algorithm, Normalization, Reinforcement Learning, MATLAB, Simulink**

I. Introduction

In recent years, the global focus on sustainability and energy efficiency has driven significant changes across various sectors, with building management standing out as a critical area of concern. Heating systems in residential and commercial buildings account for nearly 40% of global energy consumption, making them substantial contributors to both environmental degradation, through greenhouse gas emissions, and economic strain, due to high energy costs [1]. The pressing need to balance these impacts has prompted the exploration of innovative heating solutions that not only reduce energy usage but also enhance the comfort of building occupants.

The challenge is compounded by the fact that indoor climates are affected by a myriad of factors, from external temperatures and weather conditions to the structural characteristics of the building itself. As it stands now, many buildings' existing heating methods end up using more energy to compensate for the way heating is controlled. [2], [3] Specifically, in systems reliant on water-based radiator heating, the correlation between water flow and temperature introduces a scenario where the lack of responsive control mechanisms leads to energy inefficiency and occupant discomfort. This is due to the conventional systems' reliance on current temperature readings alone, neglecting the predictive insights that could inform more efficient temperature regulation [4].

In this context, reinforcement learning, particularly its application for proactive heating control, emerges as a transformative solution. By adopting a proactive stance, reinforced by the predictive power of reinforcement learning, heating systems can transcend traditional reactive models. This approach not only aims to optimize energy usage by adapting to a variety of influencing factors in real-time but also anticipates future changes, ensuring a stable and comfortable indoor environment. The exploration of reinforcement learning for heating control in this project is driven by a vision to significantly reduce energy consumption and address temperature fluctuation issues, thereby contributing to environmental sustainability while enhancing occupant comfort and well-being [5].

The primary purpose of this project is to leverage Q-learning and artificial intelligence to develop a proactive heating control system, enhancing energy efficiency and occupant comfort in buildings. By using predictive analytics and weather forecasting, the system anticipates environmental changes, adjusting heating strategies accordingly. This data-driven approach aims to reduce energy wastage and contribute to sustainable building management, aligning with broader sustainability and efficiency goals that is an active societal issue globally.

II. Background

The growing demand for improved energy efficiency in buildings has exposed the limitations of traditional heating control systems, which predominantly rely on basic thermostat feedback mechanisms. These conventional methods often fail to account for the complex variables influencing indoor climate, such as insulation quality, external weather conditions, and varying occupancy levels. As a result, they tend to consume more energy than necessary, leading to higher costs and compromised comfort for occupants. Addressing these shortcomings requires the development of more advanced, dynamic heating control systems that can adapt in real-time to these multifaceted factors [6].

In the context of the strides made in smart grid technology and building automation, which have markedly optimized

energy supply and demand dynamics, the advent of machine learning, and specifically Q-learning, stands out as a pivotal innovation [6], [7], [8]. This domain of artificial intelligence, through the lens of reinforcement learning and its subset Q-learning.

Q-learning relies on defining states, which encapsulate relevant environmental information at a given moment. Within each state, the agent has a set of actions available, forming state-action pairs. Executing an action within a state triggers a transition to a new state, altering the environment. Central to Q-learning is the construction of a Q-table, which stores evaluations for each state-action pair in the form of Q-values. These Q-values represent the expected return or utility associated with selecting a particular action in a given state. Through iterative updates based on experiences, Q-learning refines its decision-making process, ultimately yielding a Q-table that serves as a guide for future actions. It is characterized as a model-free variant of reinforcement learning, Q-learning equips heating systems with the foresight for anticipatory adjustments, thus streamlining energy efficiency and enhancing occupant comfort. It transcends traditional reactive models by employing a strategy that continuously learns and adapts from environmental interactions to optimize heating operations without the necessity for scenario-specific programming [9], [10], [11]. This dynamic adjustment capability, essential for both refining decision-making processes and achieving the highest possible rewards, highlights Q-learning's broad applicability, from automating vehicles to improving energy system efficiencies, and underscores its critical role in advancing heating control systems towards more energy-conservative and comfort-assuring operations.

Moreover, the application of Q-learning extends beyond mere temperature regulation. By integrating predictive analytics, including weather forecasting, Q-learning-equipped systems can pre-emptively adjust to expected climatic shifts, thereby avoiding the pitfalls of over- or under-heating. This not only conserves energy but also ensures a stable indoor environment, tailored to the preferences and well-being of building occupants [12].

III. Problem formulation

The challenge of balancing energy efficiency and occupant comfort in heating control systems has grown in importance amidst global sustainability efforts. Traditional systems struggle to adapt to changing conditions, resulting in energy waste and inconsistent indoor temperatures. This study aims to enhance heating control using Q-learning, integrating forecasted data to create a proactive system. Since the scope of this project is narrow, the main research questions are formulated as follows:

RQ1: What improvements in indoor temperature stability can be observed using a Q-learning algorithm implementation?

RQ2: How does a Q-learning-based heating control system impact comfort and energy efficiency in terms of cost in buildings when incorporating forecasted weather data?

IV. Related Work

In intelligent building management, advanced algorithms, particularly reinforcement learning (RL) and Q-learning, are revolutionizing HVAC operations to boost energy savings and occupant comfort. Studies in this area, notably by Park et al. [13], have successfully demonstrated the potential of RL in significantly lowering energy loads. These strategies are particularly notable for their effective control over building elements like shading and insulation, utilizing the predictive capabilities of Q-learning to anticipate and react to environmental changes, thereby optimizing energy usage and maintaining comfortable indoor conditions.

The exploration into direct policy optimization, especially through policy-based RL methods such as REINFORCE [14], marks a significant evolution in the strategies employed for HVAC control. These methods, known for their efficacy in learning stochastic policies, have expanded the control over building climate parameters beyond the discrete decision-making frameworks provided by traditional algorithms like Q-learning. By formulating more nuanced control strategies, these policy-based methods have significantly contributed to the broader goal of achieving energy-efficient and occupant-friendly environments.

Moreover, the adaptability and robustness of RL-trained policies, as investigated across various climates and seasonal changes, demonstrate an impressive capacity for maintaining optimal performance in new and unpredictable environmental conditions [15]. This adaptability is paramount for RL applications in managing the dynamic and complex scenarios often encountered in climate control, ensuring that systems can consistently optimize for both energy efficiency and occupant comfort, regardless of external variations.

There are other facets with integration of RL algorithms in the state-of-the-art with, for example, simulation platforms such as EnergyPlus which represents a substantial step forward in evaluating the practical impact of these advanced control strategies on building performance. Through detailed analyses of energy consumption, indoor temperature dynamics, and the overall effectiveness of RL-based controls, this integration provides a solid foundation for transitioning from theoretical models to real-world applications [16]. The synergy between sophisticated RL algorithms and comprehensive building simulation tools not only highlights the potential for tangible improvements in building management systems but also paves the way for the widespread adoption of intelligent climate control solutions.

V. METHOD

As detailed in previous sections, the project's scope was narrowly focused on the Q-learning algorithm, meaning there was no exploration of other methodologies such as deep deterministic stochastic gradient. The heating control system was developed by editing a publicly available model in Simulink known as "Thermal Model of a House" [17], which utilizes blocks to simulate the impact of the heating system on the building's temperature. The Q-learning algorithm was scripted in MATLAB and the Simulink environment.

A. Data Gathering

The dataset for this experiment was derived from "temperatur.nu," a weather forecasting website chosen for its comprehensive and efficient data collection, offering hourly measurements in a .csv format dating back to 2009. A Python script was utilized to filter data from a weather station in Västerås, Sweden, narrowing the scope to include only data from 23-03-2024 to 28-03-2024, based on hourly intervals. Although the raw dataset recorded multiple variables, including outside temperature, humidity, wind direction, and speed, the project focused exclusively on the outside temperature, serving as the primary variable for model training.

To augment the dataset's diversity and simulate a range of environmental conditions, two key modifications were applied: the addition of one percent Gaussian noise to the temperature data to mimic measurement variance and an adjustment of the temperature baseline by one degree in both upward and downward directions. These enhancements led to the generation of eleven unique signals, enriching the training dataset to facilitate a more robust learning experience for the model.

B. The Q-learning Algorithm

In the Q-learning implementation for controlling building temperatures, a single agent interacted with the environment. Decisions were made based on the current state of the environment and the Q-values stored in the Q-table. Actions corresponded to adjusting the change of heating or cooling systems in the building, influencing the transition to the next state.

To determine the comfort score, which reflects how close the current temperature is to the desired temperature, the formula below was used:

$$\text{CS} = \begin{cases} 1 - \left| \frac{T_{\text{indoors}} - T_{\text{set}}}{\text{max_deviation}} \right| & \text{if } |T_{\text{indoors}} - T_{\text{set}}| \leq \text{max_deviation} \\ 0 & \text{otherwise} \end{cases} \tag{1}$$

T_{set} represents the target temperature for comfort, and max_deviation is the maximum allowable deviation from this target. T_{indoors} represents the temperature indoors. Generally, the desired room temperatures is between 18°C [18]. For this it was set to 21°C. The max_deviation was set to three.

Next, the function calculates the energy score, which indicates the efficiency of the heating or cooling system. This score is determined by the ratio of the baseline heat cost to the heat cost:

$$\text{ES} = 1 - \frac{\text{BaselineHeatCost}}{\text{HeatCost}} \tag{2}$$

The energy score ranges from 0 to 1, where lower values signify higher energy usage.

Finally, the function combines the comfort score and energy score to compute the overall reward. The comfort score contributes 80% to the final reward, while the energy score contributes 20%. The weighted combination is expressed as:

$$\text{R} = \text{cs} \times 0.8 + \text{es} \times 0.2 \tag{3}$$

In the current model, the reward function within the Q-learning algorithm is calculated based on the deviation of the indoor temperature from the desired set point, with the primary focus being on maintaining this temperature close to the target value. The reward is a direct reflection of how well the action taken in a given state helps in minimizing this temperature deviation.

The reward is calculated as the negative absolute difference between the indoor temperature T_{indoors} and the set point T_{set}. This means that smaller deviations result in higher rewards, encouraging the system to take actions that keep the temperature as close to the desired level as possible. The reward function used in this model does not explicitly factor in energy efficiency; it is solely based on temperature regulation.

Regarding the training process, the Q-learning algorithm interacts with the environment iteratively over a series of episodes. Each episode represents a complete run through a predefined time period, during which the system takes actions based on the current state, observes the outcome, and updates the Q-values accordingly. The formula in the image is:

$$Q(s,a) \leftarrow Q(s,a) + \alpha[r_{ss}^{a} + \gamma \max Q(s',a') - Q(s,a)] \tag{4}$$

Where:

- $Q(s,a)$: The current Q-value for the state-action pair.
- α: The learning rate, which determines to what extent the newly acquired information will override the old information.
- $r_{ss'}$: The reward received after transitioning from state s to state s' due to action a.
- γ: The discount factor, which determines the importance of future rewards.
- $\max Q(s',a')$: The maximum Q-value for the next state s' across all possible actions a', representing the best possible future reward.

For the testing phase, the same location in Västerås was used but with a different time period—March 29th to April

3rd, 2024. This separate testing dataset, also sourced from "temperatur.nu," was utilized to evaluate the performance of the trained Q-learning model. By using data from the same location but a different time frame, the testing phase provided a robust assessment of the model's ability to generalize its learning to new but similar conditions, ensuring that the model's performance wasn't overly specific to the training data.

C. MATLAB & Simulink Environment

The "Thermal Model of a House" [17] in MATLAB Simulink simulates the operation of a home heating system, where a thermostat activates heating based on the indoor temperature's deviation from a desired set point. The system models the heat input from a heater and the loss to the external environment, factoring in the house's thermal characteristics and outdoor temperature variations. It calculates the energy consumption and cost, and a visualization component displays the indoor temperature maintenance and heating expenses over time, providing insights into the efficiency and cost-effectiveness. Figure 1 provided below is the original model.

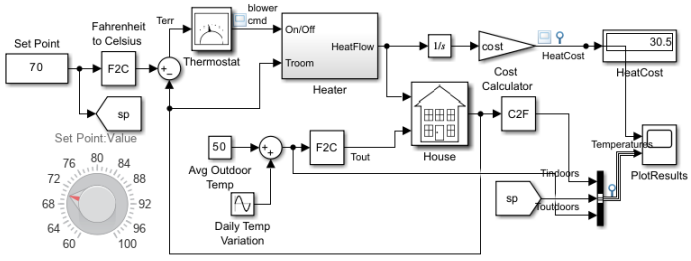


Figure 1. Thermal Model of a House - Original Model

A modified version of the original model was used as a baseline and is shown in Figure 2. The From Workspace block "Data" retrieves data from the dataset exported from temperatur.nu. Additionally, the indoor temperature is affected by the amount of heat generated by the heater. The heater operates based on signals from the thermostat, which turns the heater on when the indoor temperature drops too low and turns it off when the temperature becomes too warm.

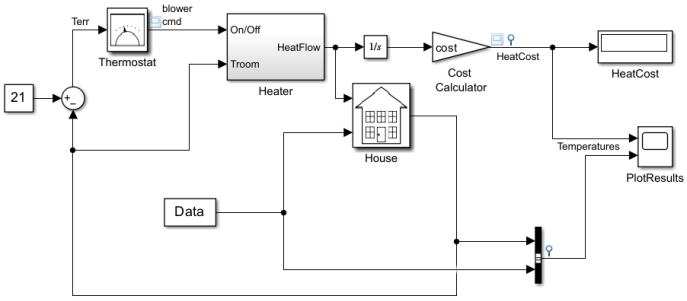


Figure 2. Modified Original Model

The model in Figure 3 uses the Q-learning algorithm as the controller for the heater, determining when it should turn on/off. The MATLAB Function block "q_learning" is the controller and handles decision-making with the heater, save states, rewards and Q-table updates. The Q-learning model operates continuously, making adjustments at each time step based on the current state and reward, rather than working through distinct episodes. This continuous learning process allows the model to adapt in real-time to the changing environment. This simulation did this over a 5-day period.

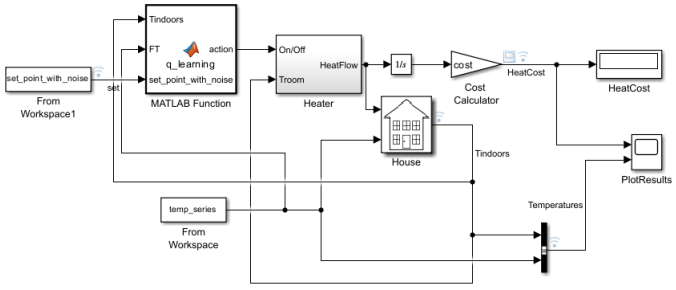


Figure 3. Proactive Q-learning model with forecast.

VI. Results

Each model was evaluated using Simulink in MATLAB, with performance visualized through MATLAB's built-in plotting tools. Figures 4 & 5 illustrate how the models adjust the heating levels, impacting the building's indoor temperature. These results were generated by testing fully trained models on the same dataset, ensuring consistency with identical training data. In the graphs, the x-axis represents time in days, while the y-axis shows temperature in Celsius. The yellow line indicates the indoor temperature, while the blue line represents the forecasted outdoor temperature.

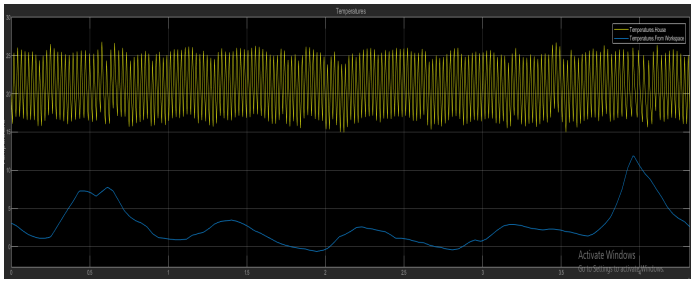


Figure 4. Graphed performance of the thermostat model (original model).

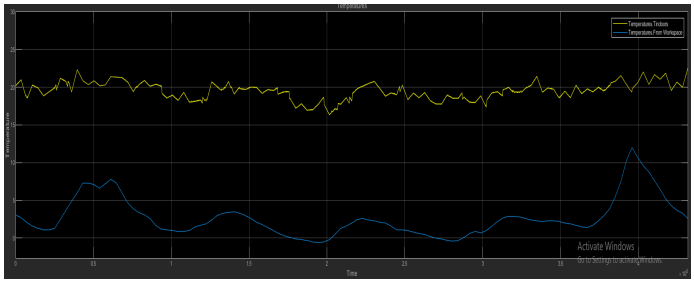


Figure 5. Graphed performance of the proactive Q-learning model with forecasted data.

The Q-learning-based heating control system was evaluated against a traditional thermostat model over a simulated period of 5 days. The evaluation focused on two primary metrics: energy efficiency & comfort level as explained in Section V, subsection B. A separate MATLAB script extracted the desired signals from both Simulink models which were: HeatCost, Set, Tindoors & Temperatures (as illustrated by a "Wi-Fi" signal in Figure 3), and thereafter analyzed.

The cumulative comfort score was calculated as the sum of deviations from the desired setpoint of 21°C over the entire simulation period. The Q-learning model, being a continuous learning system, recorded a cumulative comfort score of 42,816.00, while the original thermostat model, which operates in a discrete manner, recorded a significantly lower score of 63.15. This substantial difference is due to the continuous nature of the Q-learning model, which produces a much greater number of measurement points compared to the original model. Although the cumulative score is higher for the Q-learning model, this reflects the increased frequency of adjustments rather than a direct comparison of comfort performance between the two models.

The normalized comfort score provides a more direct comparison by accounting for the number of measurement points. As illustrated in Figure 6 the Q-learning model achieved a normalized comfort score of 0.59, whereas the thermostat model achieved a score of 3.51. The lower normalized score of the Q-learning model indicates that, on average, it maintained a temperature closer to the desired setpoint of 21°C. This demonstrates that the Q-learning model, despite its higher cumulative score, was more effective at minimizing deviations from the setpoint across its many adjustments. Based on the analysis of the normalized comfort score and a visual comparison of Figures 4 & 5, it can be concluded that the indoor temperature is significantly more stable with the Q-learning algorithm model, thereby addressing **RQ1**.

The energy efficiency of the heating was reflected by the cost associated with maintaining the desired indoor temperature. The Q-learning model demonstrated an energy cost of $116.2, which is lower than the original thermostat model's cost of $128.4 by $12.2. (9.5%). This indicates that the Q-learning model not only improved temperature stability but also reduced the energy consumption required to maintain this stability. This is illustrated in Figure 7. Considering the analysis of energy efficiency, cost differences, and comfort scores presented earlier, thereby addressing **RQ2**.

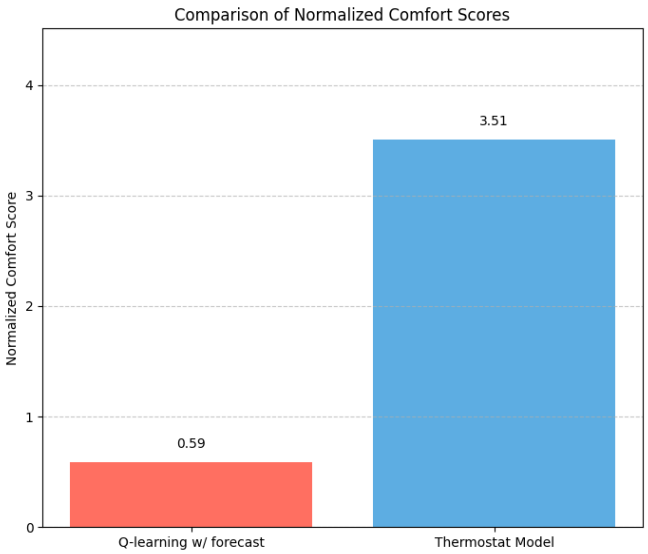


Figure 6. Results comparing the normalized comfort scores between the thermostat model and the Q-learning model.

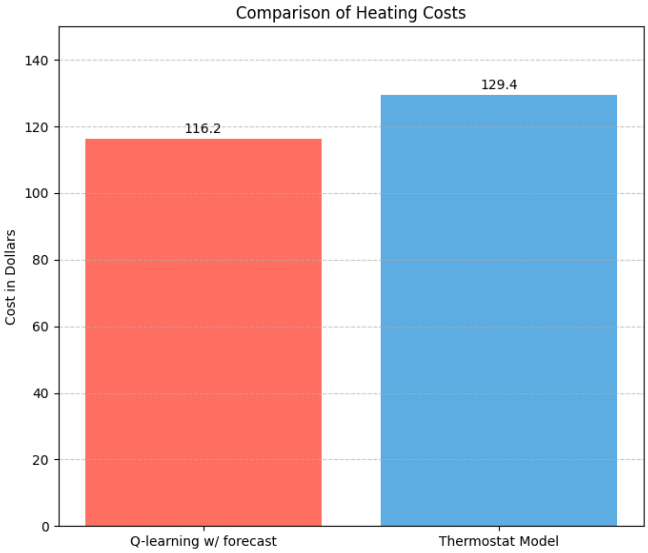


Figure 7. Results comparing the heating costs between the thermostat model and the Q-learning model.

VII. Discussion & Conclusions

The implementation of a Q-learning algorithm for heating control, utilizing forecasted weather data, demonstrated significant improvements in energy efficiency and cost-effectiveness compared to the original thermostat model. The Q-learning model achieved a heating cost of $116.2, which is approximately 9.5% lower than the cost of the thermostat model ($128.4). This reduction highlights the algorithm's ability to optimize energy usage by making more informed decisions based on both current indoor temperatures and forecasted outdoor conditions.

The analysis of the normalized comfort score further supports the effectiveness of the Q-learning approach. The continuous learning nature of the Q-learning algorithm allows for finer adjustments to the heating system, resulting in a more stable indoor temperature, as evidenced by the comparison of temperature fluctuations in the visualized data. Despite the higher cumulative comfort score due to the increased frequency of adjustments in the Q-learning model, the normalized score indicates that the comfort level remains within acceptable bounds, ensuring that the system is responsive without compromising occupant comfort.

The results highlight the need to carefully balance exploration and exploitation within the Q-learning framework. While frequent adjustments help maintain temperature stability, they could potentially increase wear on heating system components over time. Future research could focus on refining this balance to enhance long-term system performance and durability. Additionally, investigating other AI approaches, such as deep reinforcement learning, might offer more efficient alternatives to the Q-learning method. Furthermore, the forecasted data used in this study had limited variability. Utilizing a larger dataset with greater temperature fluctuations could make the heating control system more robust and better equipped to handle complex environmental conditions.

One limitation of this study is that it tested only two models: the proactive Q-learning model and the traditional thermostat model. Including a third, retroactive Q-learning model that does not utilize forecast data could offer additional insights into which approach is more effective. Future work should consider incorporating this third model to gain a more complete understanding of how different strategies affect both energy efficiency and indoor comfort.

In conclusion, the Q-learning algorithm with integrated weather forecasts outperformed the traditional thermostat model in terms of energy efficiency and cost savings, while maintaining a stable and comfortable indoor environment. These findings suggest that adaptive, learning-based control systems have significant potential for improving the efficiency of residential heating systems, particularly in the face of varying and unpredictable weather conditions. Further research could extend this work by exploring the application of similar algorithms to other aspects of home energy management, as well as investigating the long-term impacts of continuous learning on system maintenance and occupant satisfaction.

VIII. Acknowledgement

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